

Yan Zhu

Irvine, California

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Education

University of Wisconsin–Madison

Madison, WI

M.S. IN PSYCHOLOGY, DATA SCIENCE IN HUMAN BEHAVIOR

Dec. 2023

- Selected Coursework: Design and Analysis of Psychological Experiments I–II; Programming in Human Behavioral Data Science; Applied Machine Learning for Behavioral Data Science; Large-Scale Behavioral Data Science; Information Visualization.

Michigan State University

East Lansing, MI

B.S. IN PSYCHOLOGY, WITH HONOR

Aug. 2022

- Minors: Youth and Society; Philosophy.
- Selected Coursework: Computational Modeling and Data Analysis I–II; Data Analysis in Psychological Research; Design and Measurement; Ecological Perspectives on Adolescents; Substance Abuse; Child & Family Psychopathology.

Seattle Central College

Seattle, WA

A.A. IN PSYCHOLOGY

2020

- Selected Coursework: Fundamentals of Psychological Research; Lifespan Psychology; Physiological Psychology.

Professional Training

University of California, Santa Barbara – Professional and Continuing Education

Santa Barbara, CA

APPLIED BEHAVIOR ANALYSIS PROFESSIONAL COURSEWORK

2025 – Present

- Selected Coursework: Single Case Experimental Design; Procedures, Interventions, and Supervision; Cognitive Development in Autism & Other Severe Disabilities; Behavior Assessment & Intervention for Children & Adolescents.

Research Experience

From Repertoire to Implementation

Irvine, CA

INDEPENDENT RESEARCHER

Jun. 2026 – Present

- Designed and conducted an independent secondary analysis of longitudinal daily-diary data to examine how protective behavioral strategies (PBS) translate from baseline repertoire into implementation during real-world drinking occasions.
- Developed a reproducible R workflow for data cleaning, scale construction, person-mean decomposition, and statistical modeling; evaluated count, ordinal, and binary outcomes using Poisson/COM-Poisson and mixed-effects regression models with diagnostic and sensitivity analyses.
- Preliminary analyses indicated that stronger baseline PBS repertoire predicted greater strategy implementation, while day-level impulsivity and psychological distress did not consistently explain within-person variation in implementation; moderation by baseline emotion-regulation difficulties was not robustly supported.
- Extended the analysis to behavioral outcomes; preliminary results suggested that greater occasion-level PBS implementation was associated with lower alcohol consumption and lower likelihood of blackout.

Addiction Research Center, University of Wisconsin–Madison

Madison, WI

GRADUATE RESEARCH ASSISTANT

Jan. 2024 – Aug. 2024

- Managed and analyzed 70,000+ ecological momentary assessment observations from 336 participants in a study of behavioral predictors of alcohol relapse.
- Conducted exploratory data analysis and data-quality validation in R to identify behavioral patterns and support the development of an alcohol-relapse prediction model.
- Developed reproducible HTML reports documenting data checks, analyses, and findings to support research interpretation and transparency.

Preventing Interpersonal Violence and Overcoming Trauma Lab, University of Wisconsin–Madison

Madison, WI

GRADUATE RESEARCH ASSISTANT

Sep. 2023 – May 2024

- Supported trauma-related research through data cleaning, variable construction, exploratory data analysis, and statistical analysis in R.
- Developed exploratory visualizations to identify and communicate patterns in behavioral and survey data.
- Created and refined funnel plots and other meta-analytic visualizations for a doctoral research project.

Healthy Minds Innovations – Graduate Data Science Capstone

Madison, WI

DATA SCIENCE CAPSTONE INTERN

May 2023 – Dec. 2023

- Conducted a graduate capstone project using 5GB+ of application and survey data to examine how users engaged with a digital mindfulness intervention in real-world use.
- Cleaned and structured complex behavioral data and applied process-mining methods in Python using PM4Py to characterize user engagement pathways.
- Examined associations between meditation engagement, including frequency and duration, and mental health and well-being outcomes using statistical and predictive modeling.

Methods & Technical Skills

Quantitative Methods: Longitudinal and repeated-measures analysis; multilevel and mixed-effects modeling; generalized linear models for count, ordinal, and binary outcomes; within-/between-person decomposition; predictive modeling.

Research & Data: Exploratory data analysis; data cleaning and transformation; scale and composite-score construction; model diagnostics and sensitivity analysis; statistical visualization; reproducible research and reporting.

Programming & Tools: R; Python; SQL; Git/GitHub; Jupyter Notebook; R Markdown; LaTeX.

Applied Behavioral Experience

LEARN Behavioral

Madison, WI / Irvine, CA

ADVANCED BEHAVIOR TECHNICIAN (FORMERLY BEHAVIOR TECHNICIAN)

Jun. 2024 – Present

- Provide direct ABA-based intervention to children with autism and developmental disabilities, working closely with clients and families under BCBA supervision.
- Implement interventions across center-based, home, and community settings, gaining direct experience with how behavioral skills and intervention strategies generalize across caregivers, routines, and structured and naturalistic environments.
- Collect behavioral and skill-acquisition data to monitor individual progress and inform ongoing treatment implementation.
- Work directly with caregivers and clinical teams, developing firsthand exposure to variability in family engagement, implementation context, and children's responses to intervention.

Honors & Awards

2020–2022 **Dean's List**, Michigan State University

2020–2021 **Spartan Transfer Scholarship**, Michigan State University

\$1,500/year

2020–2021 **Associate Achievement Award**, Michigan State University

\$1,500/year

Service & Volunteer Experience

Canopy Center

Madison, WI

FAMILY INTERACTION SPECIALIST, PARENT TO CHILD (P2C) PROGRAM

Jun. 2023 – Sep. 2023

- Supervised court-ordered parent-child visits and documented objective observations of family interactions, contributing 64 volunteer hours to a Dane County Human Services initiative.
- Independently analyzed volunteer demographic data to identify patterns relevant to recruitment and engagement and communicated findings to HR.
- Conducted exploratory analyses of case duration and service frequency to support internal program evaluation.